



Editor's choice

The kinetics of the reactions of ground-state NH with N₂O and implications for ammonia combustion

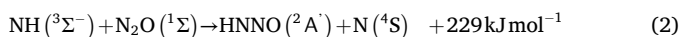
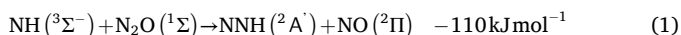
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ABSTRACT

Quantum chemistry and canonical transition state theory have been applied to derive rate constants k for reactions of $^3\Sigma^-$ imidogen with nitrous oxide. 3 channels were quantified. The fastest leads to $\text{NNH} + \text{NO}$ and may be summarized as $k = 2.8 \times 10^7 \text{ T}^{1.68} \exp(-23.1 \text{ kcal mol}^{-1}/\text{RT}) \text{ cm}^3 \text{ mole}^{-1} \text{ s}^{-1}$. Two much slower pathways lead to $\text{N}_2 + ^3\text{HNO}$ and spin-forbidden ^1HNO . Modeling of literature data, from an NH_3/O_2 flame and jet-stirred reactor measurements on dilute $\text{NH}_3/\text{N}_2\text{O}$ mixtures, with the new kinetic information suggests that these channels are too slow to consume N_2O significantly, and confirms that a widely used proposed rate constant for $\text{NH} + \text{N}_2\text{O}$ overestimates the reactivity.

The possible use of ammonia (NH_3) as a carbon-free energy transfer fuel has received much attention [1–3]. One potential problem is the formation of nitrogen oxides as unwanted products, and the issue of nitrous oxide (N_2O) release is of special interest because it is a greenhouse gas with a global warming potential about 270 times greater than for carbon dioxide over 100 years [4]. Thus modest N_2O emissions could negate the advantage of NH_3 as a zero- CO_2 fuel [5]. There has been a renewed focus on high-temperature radical chemistry that could destroy N_2O during combustion, for instance via reaction with oxygen atoms [6,7], hydrogen atoms [8] and, of particular interest in NH_3 flames, NH_2 radicals [9,10]. A key radical in ammonia flames is imidogen (NH). Hanson and Salimian [11] proposed an activation energy for $\text{NH} + \text{N}_2\text{O}$ of 25 kJ mol^{-1} so there is potential for high reactivity at elevated temperatures. In a recent high-temperature modeling study of $\text{NH}_3/\text{N}_2\text{O}$ (and NO) mixtures by Zou et al., the authors employed this estimate for $\text{NH} + \text{N}_2\text{O}$ and noted it as a critical target for further evaluation [12].

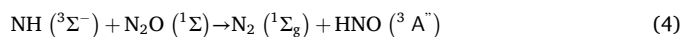
Ground state NH is a triplet molecule in the $^3\Sigma^-$ state and the lowest-lying singlet state is $^1\Delta$. We envisage four reactions of nitrous oxide, shown here with the reaction enthalpy at 298 K based on data from the Active Thermochemical Tables [13], except for HNNO taken from Glarborg et al. [8]:



Both these reactions are spin-allowed, and (1) is the analog of the channel for isoelectronic $\text{O}(^3\text{P}) + \text{N}_2\text{O} \rightarrow \text{NO} + \text{NO}$. The high endothermicity of (2) means this process will be too slow to be of practical importance and we do not pursue it further. Continuing the isoelectronic analogy with $\text{O} + \text{N}_2\text{O}$, we consider.



which is highly exothermic. However, the analogy breaks down when we note that ground-state HNO is a singlet, unlike the analogous $\text{O}_2(^3\Sigma_g^-)$ product of $\text{O}(^3\text{P}) + \text{N}_2\text{O}$. Therefore spin-forbidden (3) must involve intersystem crossing (ISC). In 1984 Hanson and Salimian estimated $k_3 = 3.3 \times 10^{-12} \exp(-24.9 \text{ kJ mol}^{-1}/\text{RT}) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ [11]. 40 years later, Wang et al. [10] needed to reduce the pre-exponential parameter by a factor of 10^4 to match observed concentration profiles, so there is a large discrepancy to resolve for (3). The spin-allowed reaction.



that creates HNO in an excited electronic state is still highly exothermic given that the singlet-triplet excitation energy is estimated here as ca. 93 kJ mol^{-1} . We expect that triplet HNO will undergo internal conversion or collisional quenching to end ultimately in the singlet state, so that (4) offers an alternative pathway for N_2O destruction. We also considered.

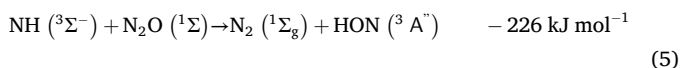
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which is exothermic and spin-allowed, but were unable to find a single-collision reaction path leading to hydroxy nitrene.

There are no direct experimental data for the bimolecular rate constants $k_{1,3,4}$ so we have applied quantum chemistry methods. We then

employ the results in multireaction modeling to compare with literature data for systems where NH might be important, to evaluate the roles of channels (1), (3) and (4).

Our first step was to obtain geometries of reactants, transition states (TSs) and products with the B2PLYP-D3 functional [14,15] and the cc-pVTZ basis set [16]. This choice was based on Klippenstein et al.'s observation that this functional yields geometries and harmonic

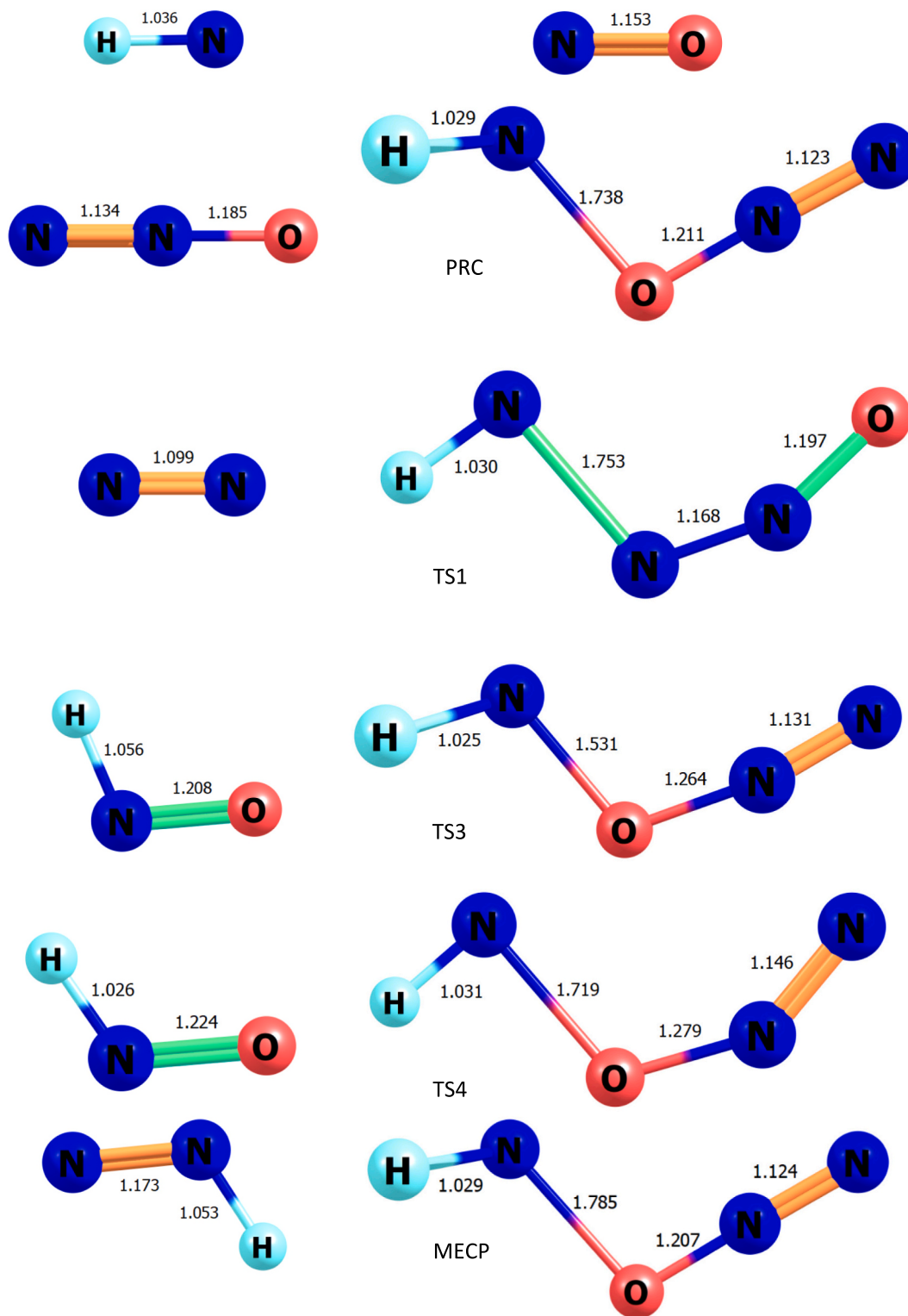


Fig. 1. Structures of stationary points for $\text{NH} + \text{N}_2\text{O}$ reactions computed with B2PLYP-D3/cc-pVTZ theory. The colour scheme in the online version is H pale blue, N dark blue, O red. Bond lengths are shown in units of 10^{-10} m. See Table S1 for cartesian coordinates. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

frequencies that compare favorably with CCSD(T) results, and that the geometries yield lower energies than CCSD(T) structures for CCSD(T)-F12 energies.[17] We used the Gaussian16 program [18] for these density functional theory calculations. The structures of the stationary points are shown in Fig. 1, while full cartesian coordinates are provided in the Supplementary Material (Table S1).

At each of the geometries, harmonic vibrational frequencies were obtained to check the nature of the stationary points, followed by anharmonic frequency calculations made with second-order vibrational perturbation theory (VPT2) and the HDCPT2 option for hybrid degeneracy corrections to resonances [19]. This yielded the anharmonic zero-point vibrational energy, E_{zpe} , and fundamental frequencies later used in the kinetic calculations. Next, the Molpro program [20,21] was used to deduce the energy via coupled cluster theory with single, double and iterative triple substitutions (CCSD(T)) at the complete basis set (CBS) limit, $E_{\text{CCSD(T)}}$. A $1/n^3$ extrapolation of the valence correlation energy with aug-cc-pVTZ and aug-cc-pVQZ basis sets was added to the HF/aug-cc-pVQZ energy. An energy correction E_{CV} for the influence of core electron correlation was derived as the difference between inclusion or exclusion of core orbitals with the cc-pwCVTZ and cc-pwCVQZ basis sets, extrapolated to the CCSD(T)/CBS limit. The latter basis was also employed to calculate a relativistic correction E_{rel} . All these single-point calculations were made with spin-restricted RHF wavefunctions and spin-unrestricted CCSD(T), and the T_1 diagnostic [22,23] was monitored to check for significant multireference character (see Supplementary Information. The open-shell singlet character of PRC and TS3 is discussed there). The last correction, E_{Q} , for correlation beyond CCSD(T) was made as the difference between unrestricted CCSDT(Q) and CCSD(T) energies based on an unrestricted HF/cc-pVDZ reference, computed using the Cfour [24] and MRCC [25] programs. This E_{Q} term partly offsets the impact of multireference character. The total energy was obtained as

$$E_{\text{tot}} = E_{\text{CCSD(T)}} + E_{\text{zpe}} + E_{\text{CV}} + E_{\text{rel}} + E_{\text{Q}} \quad (6)$$

Our methodology is similar in spirit to, for example, the ANL0 composite method [17] but with simplifications. Differences in E_{tot} correspond physically to enthalpy changes at 0 K, ΔH_0 .

Components of the energy calculations are tabulated in Table S2, while derived 0 K enthalpies relative to the reactants are listed in Table 1. As a check on the accuracy of these values, comparison with the Active Thermochemical Tables [13] indicates errors in ΔH_0 for reactions 1 and 3 of -0.6 and $+0.7$ kJ mol $^{-1}$, respectively.

Fig. 2 shows a potential energy diagram for NH + N₂O. Reaction (1) has the lowest barrier and is the dominant pathway. The more exothermic pathways (3) and (4) have barriers higher by about 60 kJ mol $^{-1}$ and therefore will be considerably slower than (1).

The most exothermic channel (3) is spin-forbidden and passes through a singlet transition state. This TS3 is the bottleneck for the reaction NH ($^1\Delta$) + N₂O. For the analogous interaction O (1D) + N₂O there is no barrier to N₂ + O₂ formation, which means this channel might be accessible to ISC where a rising triplet potential energy curve for O (3P) + N₂O crosses a falling singlet O (1D) + N₂O curve. We investigated

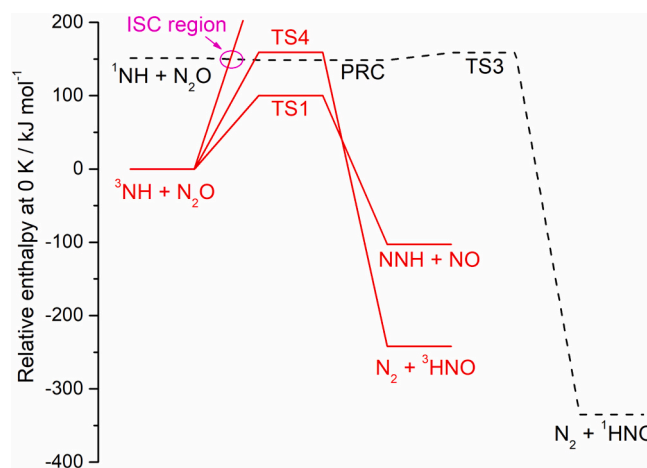


Fig. 2. Potential energy for NH + N₂O reactions (see text) including zero-point vibrational energy. The energy of NH ($^1\Delta$) relative to the ground state is taken from experiment [26], and the singlet-triplet MECP is located near the singlet prereaction complex (see text). Triplet paths are shown as solid red lines, and the singlet path is shown as a dashed black line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

energy changes as NH approaches the O-atom of N₂O using B2PLYP-D3/cc-pVTZ density functional calculations implemented with the Orca program [27,28]. By contrast to the singlet surface for O + N₂O, we find (A) that there is a weakly bound singlet HN-ONN prereaction complex (PRC), with $r(\text{N}-\text{O}) = 1.80 \times 10^{-10}$ m, before TS3 (where $r(\text{N}-\text{O}) = 1.46 \times 10^{-10}$ m), and (B) that the MECP (see Fig. 1 and the Supplementary Material for more details) with the triplet surface occurs at $r(\text{N}-\text{O}) = 1.78 \times 10^{-10}$ m, close to the PRC. This means that triplet trajectories of NH with N₂O that undergo ISC must still pass through TS3. The possibility of long-lived trajectories in the PRC may provide extended opportunities for ISC. At the high-pressure limit (HPL) the PRC is in equilibrium with the reactants (equilibrium constant K_{eq}) and then the probability of ISC does not enter into the kinetics. This is an upper bound to k_3 because at sufficiently low pressures, collisions may not maintain equilibrium and the kinetics become sensitive to the perhaps small ISC probability. Thus $k_3(\text{HPL}) = K_{\text{eq}}(\text{singlet PRC formation from } ^3\text{NH} + \text{N}_2\text{O}) \times k(\text{PRC dissociation through TS3})$. Equivalently, reaction (3) at the HPL can be modeled as the product $K_{\text{eq}}(^3\text{NH} \rightleftharpoons ^1\text{NH}) \times k(^1\text{NH} + \text{N}_2\text{O} \text{ reaction through TS3})$. The small values for k_3 means that overall removal of N₂O is insensitive to details of the analysis of the spin-forbidden pathway.

Rate constants were evaluated via canonical transition state theory using the Multiwell program [29], with tunneling implemented via the Eckart model. Each of the transition states involved a low-barrier torsion. These were characterized with B2PLYP-D3/cc-pVTZ theory for 32 points along a 2π relaxed scan of the torsional angle, then a discrete Fourier Transform yielded cosine and sine coefficients for use in Multiwell's solver for partition functions of torsional potentials. The results for the three channels considered are plotted in Arrhenius form in Fig. 3. As noted above, reaction (1) leading to NNH + NO is the fastest process. Nevertheless, it is approximately 3 times slower than the analogous reaction O (3P) + N₂O $\rightarrow$ NO + NO with a similar activation energy. The difference comes from the greater loss of entropy when NH is bound in the TS. By chance, TS3 and TS4 have very similar energies (see Table 1), but $k_4 > k_3(\text{HPL})$ in part because of the greater electronic degeneracy of TS4 compared to TS3. The present results support a significant lowering of the NH + N₂O reactivity employed in flame models, compared to the estimate by Hanson and Salimian [11]. Here we argue for high activation energies, while Wang et al. accomplished lowering of k_3 via a pragmatic reduction of its pre-exponential factor by 10^4 . The question then arises, are any of the reaction channels still significant?

Table 1

Computed enthalpies at 0 K relative to NH ($^3\Sigma^-$) + N₂O ($^1\Sigma$).

Species	$H_{\text{rel}} / \text{kJ mol}^{-1}$
NH ($^3\Sigma^-$) + N ₂ O ($^1\Sigma$)	0
NH ($^1\Delta$) + N ₂ O ($^1\Sigma$)	151.4 (expt [26])
NNH ($^2A'$) + NO ($^2\Pi$)	-102.8
N ₂ ($^1\Sigma_g$) + HNO ($^1A'$)	-335.2
N ₂ ($^1\Sigma_g$) + HNO ($^3A'$)	-242.1
N ₂ ($^1\Sigma_g$) + HON ($^3A'$)	-226.6
PRC	148.7
TS1	100.2
TS3	158.8
TS4	159.0

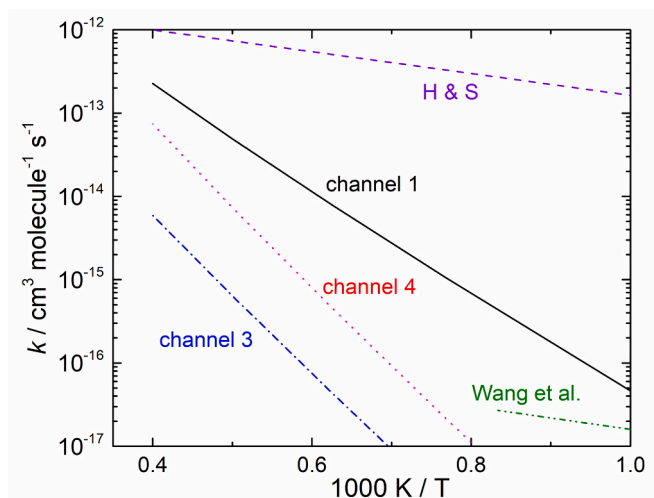


Fig. 3. Arrhenius plot of rate constants k computed for channels 1, 3 and 4 (see text), along with the estimate by Hanson and Salimian [11] and the values chosen by Wang et al. [10].

In ammonia flames, [NH] can become larger than [O] under stoichiometric or reducing conditions, so the $\text{NH} + \text{N}_2\text{O}$ reactions may contribute to limiting nitrous oxide emissions from combustion of ammonia. In this section, the impact of the reaction will be investigated for NH_3 oxidation by N_2O and O_2 , respectively. The chemical kinetic model is adopted from the recent study of NH_3 oxidation by Jian et al. [30], with the N_2O chemistry subset updated according to results for $\text{N}_2\text{O} + \text{O}$ and $\text{N}_2\text{O} + \text{H}$ [6,8].

We added the $\text{NH} + \text{N}_2\text{O}$ kinetics as

$$k_1 = 2.8 \times 10^7 T^{1.68} \exp(-23.1 \text{ kcal mol}^{-1}/RT) \text{ cm}^3 \text{ mole}^{-1} \text{ s}^{-1} \quad (7)$$

where rate expressions were evaluated at up to 2500 K. Similarly, the minor pathways were determined to be.

$$k_3 = 4.6 \times 10^6 T^{1.78} \exp(-36.3 \text{ kcal mol}^{-1}/RT) \text{ cm}^3 \text{ mole}^{-1} \text{ s}^{-1} \quad (8)$$

$$k_4 = 2.1 \times 10^6 T^{2.21} \exp(-36.2 \text{ kcal mol}^{-1}/RT) \text{ cm}^3 \text{ mole}^{-1} \text{ s}^{-1} \quad (9)$$

Fig. 4 shows a comparison of modeling predictions with experimental results from Wang et al. [10] for NO in the reaction of a dilute mixture of NH_3 and N_2O in a jet-stirred reactor. NO formation provides a measure of the importance of total $\text{NH} + \text{N}_2\text{O}$ reactivity, since either NO or HNO is produced and HNO is then largely converted to NO. With the present values of k_1 , k_3 and k_4 , $\text{NH} + \text{N}_2\text{O}$ is too slow to contribute to NO formation. The good agreement with experiment supports this finding. On the other hand, use of the estimate from Hanson and Salimian [11] for $\text{NH} + \text{N}_2\text{O} \rightarrow \text{N}_2 + \text{HNO}$ yields overprediction of NO. This is consistent with the analysis of Wang et al. [10], who suggested that the estimated rate constant from Hanson and Salimian should be reduced by four orders of magnitude.

Low-pressure flame data of Bian et al. [31] are compared with modeling results in Fig. 5. Bian et al. measured the structure of a flame burning in a near equimolar ammonia-oxygen mixture at reduced pressure (35 Torr), using a molecular beam sampling technique coupled with a mass spectrometer. In the figure, results are shown for NH_2 , NH , and N_2O .

The predicted peak values of NH_2 , NH , and N_2O agree with the measurements within their uncertainty, but for NH_2 and N_2O the profiles are broader than observed. Also, the predicted NH peak is shifted downstream, compared to experiment. The lines for results with and without the $\text{NH} + \text{N}_2\text{O}$ reactions are essentially the same (superimposed), which shows that the $\text{NH} + \text{N}_2\text{O}$ reactions do not influence the predicted profiles significantly. Under the conditions in the flame, this

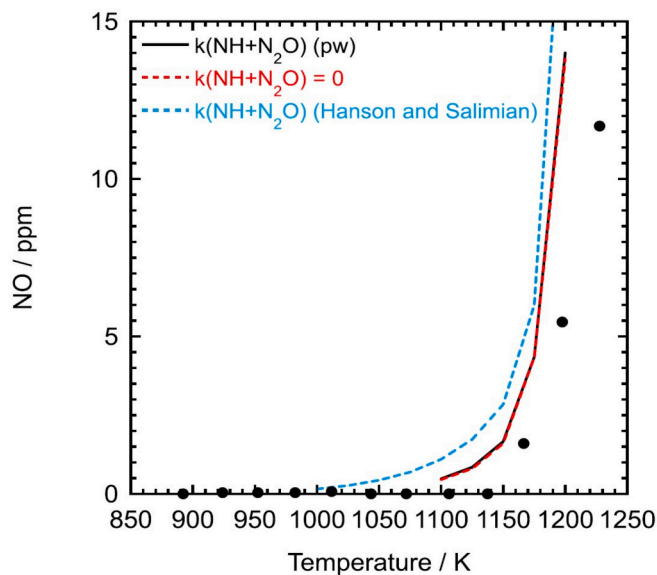


Fig. 4. Comparison of modeling predictions with the experimental data from Wang et al. [10] for NO in reaction of a dilute mixture of NH_3 and N_2O in a jet-stirred reactor. Symbols denote experimental data, while lines denote modeling predictions with the present mechanism, with the $\text{NH} + \text{N}_2\text{O}$ reaction rate constant taken from the present work (solid line), set to zero (red dashed line), or drawn from Hanson and Salimian [11] (blue dashed line). Inlet concentrations: 0.25 % NH_3 , 1.0 % N_2O ; Ar balance. The pressure is atmospheric and the residence time $\tau = 2$ s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

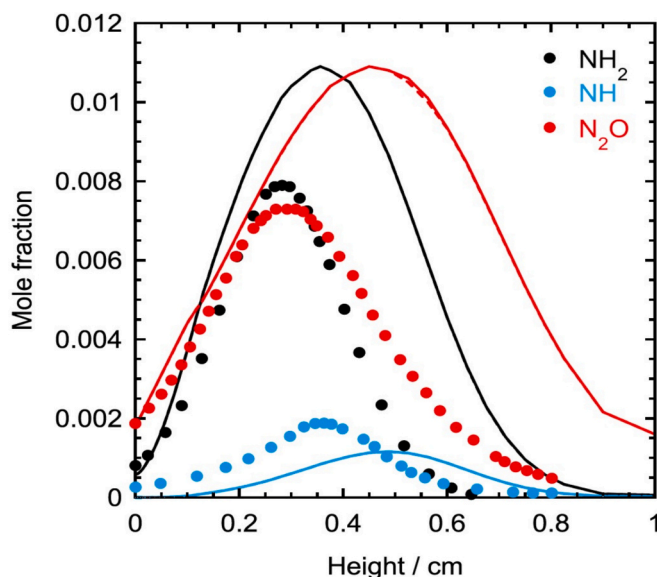


Fig. 5. Comparison of measured [31] and predicted mole fractions of NH_2 , NH , and N_2O in a low-pressure premixed ammonia/oxygen flame. The symbols denote experimental data, while the solid and dashed lines denote calculated values with and without the $\text{NH} + \text{N}_2\text{O}$ reaction. Conditions: pressure of 35 Torr; inlet composition 48 % NH_3 , 51 % O_2 , and 1 % Ar. The gas velocity was 60.8 cm s^{-1} and the peak flame temperature was about 2250 K.

chemistry is too slow to compete with other consumption steps for N_2O , primarily thermal dissociation and reaction with atomic hydrogen. Conditions that favor formation of NH , i.e., high temperature and limited oxygen availability, also tend to promote the $\text{N}_2\text{O} + \text{M}$ dissociation and $\text{N}_2\text{O} + \text{H}$ reactions.

The present work indicates that the overall reaction of NH with N_2O is too slow to reduce emissions of nitrous oxide in ammonia combustion.

The $\text{NH} + \text{N}_2\text{O}$ reactions have significant barriers, limiting their importance at medium temperatures where thermal dissociation of N_2O is slow. Under oxidizing conditions and medium temperatures, where even the similar $\text{N}_2\text{O} + \text{O}$ reaction becomes competitive, formation of NH is at too small a concentration for $\text{NH} + \text{N}_2\text{O}$ to play a role.

In conclusion, we have evaluated kinetics for NH ($^3\Sigma^-$) reactions with N_2O for three pathways, derived via a quantum chemistry/transition state theory approach. The main channel is spin-allowed formation of $\text{NO} + \text{NNH}$, with a rate constant approximately one third of that for the analogous process $\text{O} (^3\text{P}) + \text{N}_2\text{O} \rightarrow \text{NO} + \text{NO}$. Numerical analysis of literature data suggests that $\text{NH} + \text{N}_2\text{O}$ chemistry is too slow, because of significant barriers, to consume N_2O significantly.

CRediT authorship contribution statement

Paul Marshall: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Peter Glarborg:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cplett.2025.142307>.

Data availability

Data will be made available on request.

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